Gulati 2020 Breast — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Four samples (≥100 malignant cells) intermix per-neighbourhood on one continuous manifold — iLISI rises 2.77→3.56 of 4 after Harmony (θ=5), one connected component, no islands removed. Read on the regenerated embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

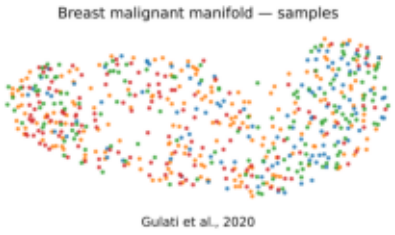
A single (Gast/pSom)→Crest developmental axis. Gast/pSom do NOT diverge into cross-repressing arms (raw r=−0.05), while both oppose Crest (Gast–Crest r=−0.50, pSom–Crest r=−0.60). Crest is the inverse terminal readout (rises along pseudotime ρ=+0.93; NC-high decile at pt 1.00; joint pSom&Crest enrichment 0.31<1). Non-branching case → GO.

Gate 3 — hub cross-repression → NA

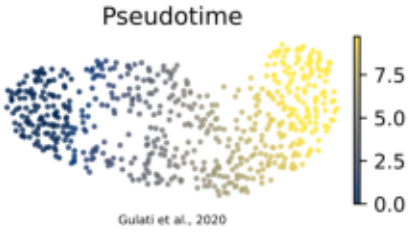
No bifurcation (curve: 2 tips / 0 forks), so there is no Mixed hub to test for Gast↔pSom anti-correlation. Branching-only → NA; 04_correlation / 05_modules not produced.

STEP FIGURES

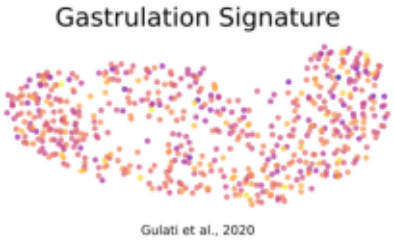
Manifold — samples



Pseudotime



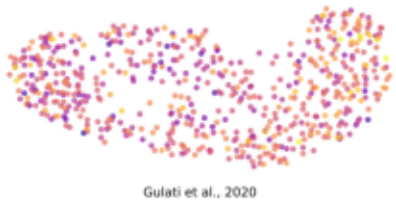
Gastrulation



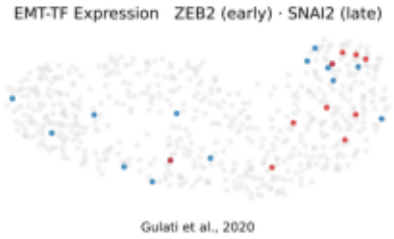
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Gulati et al., 2020), SmartSeq2 — 861 malignant cells (8 samples) → 587 analysed (4 samples ≥100); single low-cycling Body (no proliferative compartment scoped out).

Verdict.
Pattern reproduces (non-branching, Linear-pSom/Crest): a single Gastrulation/Somitogenesis → Neural-Crest developmental axis with Crest as the inverse terminal readout; no Gast↔pSom bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (root-end) → SNAI2 (terminus).

Parameter choices

Malignant / threshold	861 → floor ≥100 (counts smooth, no cliff) → 4 samples / 587 cells; 0 islands
Trajectory / geometry	scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips / 0 forks; SA 2.89
Subtype / annotation	single low-cycling Body (no proliferative compartment scoped out); cell_type = Malignant
Expression / integration	SmartSeq2 TPM (log1p); overdisp=98; n_pcs=20; Harmony(sample) θ=5; k=30; spread=1.3
Root	curve tip 0 (Gast/pSom-high, Crest-low origin); Crest terminal; pt vs prolif ρ=−0.01
EMT-TFs	ZEB2 (root-end) / SNAI2 (terminus) from 10-TF panel; detection-gated; VIM excluded